

VR-based space exploration system (VRSES)

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Abstract—Astronomy education presents a unique challenge: the subjects of study are vast, distant, and entirely inaccessible through conventional laboratory methods. Students learning about extrasolar star systems have limited means of developing spatial intuition and experiential understanding beyond two-dimensional textbooks and static simulations. Virtual reality (VR) is an emerging technology with significant potential for bridging this gap. In this paper, an immersive VR-based space exploration system (VRSES) is developed, which enables users to navigate and explore a hypothetical binary star system in a fully interactive three-dimensional environment. Users can observe the host stars, diverse planetary bodies, and complex orbital mechanics from multiple vantage points, fostering a more intuitive grasp of astronomical scale and structure. The system is implemented using the Unity 3D game engine with C# scripting, targeting standard VR-compatible hardware. Rather than relying on standard solar system data, the proposed system employs a learner-centred design that immerses users in a custom-designed, multi-body gravitational environment. By navigating correctly scaled stellar relationships and procedurally generated atmospheric conditions, the application aims to reduce extraneous cognitive load and promote deep conceptual schema formation. Ultimately, this project details the system's architecture and interactive elements, demonstrating how immersive VR environments hold strong promise as scalable pedagogical tools to enhance science education.

Index Terms—virtual reality, space exploration, astronomy education, extrasolar systems, immersive learning, Unity 3D, cognitive load

I. INTRODUCTION

Astronomy occupies a unique position in science education: it demands students comprehend phenomena that operate at scales entirely beyond human perception. Unlike chemistry or electronics, where students can handle physical equipment and directly observe reactions, astronomical learning has traditionally been confined to diagrams, photographs, and abstract mathematical descriptions. The concept of extrasolar star systems—planetary bodies orbiting stars beyond our own Sun—is particularly abstract, relying on the student's ability to mentally construct three-dimensional spatial relationships from two-dimensional representations.

The limitations of traditional instructional modalities in astronomy are well documented. Static imagery fails to communicate the relative scale and three-dimensional geometry of star systems, while conventional planetarium software, though useful, typically offers a passive observation experience rather than interactive exploration. As a result, students often leave astronomy courses with significant misconceptions about orbital geometry, stellar distances, and the comparative scale of celestial bodies.

Virtual reality (VR) technology offers a compelling solution. By placing users inside a fully rendered three-dimensional environment, VR enables a form of embodied learning that is qualitatively different from observing a screen. Prior research has demonstrated VR's effectiveness in domains ranging from medical training to architectural visualization and molecular chemistry education. However, the application of immersive VR specifically to multi-body extrasolar astronomical exploration remains relatively underexplored in the educational technology literature.

This paper presents the design and development of a VR-based Space Exploration System (VRSES), a Unity 3D application that enables users to navigate interactively through a custom-designed, hypothetical extrasolar environment. The system models an imaginary binary star system, rendering the host stars, planetary bodies, and complex orbital paths in three dimensions. Users can alter their viewpoint, scale their perspective from planetary to stellar distances, and interact with informational overlays. By utilizing a procedurally generated multi-body model rather than standard solar system data, the environment forces learners to build new spatial intuitions, expanding their astronomical understanding.

The remainder of this paper is organized as follows. Section II details the core project objectives and the development methodology. Section (III) presents the system architecture and implementation, covering the 3D modeling, procedural scene construction, and VR interaction design. Finally, Section (IV) concludes the paper with a discussion on the system's pedagogical value, personal reflections, and potential future

applications.

A. Motivation and Background

The cognitive challenge of learning extrasolar astronomy is substantially extrinsic in nature. According to cognitive load theory, the total mental effort expended during learning is the sum of intrinsic load (arising from the inherent complexity of the subject matter), extraneous load (arising from the design of instructional materials), and germane load (devoted to schema formation and deep learning). In traditional astronomy instruction, the extraneous cognitive load is high: students must mentally translate two-dimensional representations into three-dimensional spatial models while simultaneously processing complex concepts like multi-body gravity and orbital dynamics. This dual demand on working memory impedes genuine conceptual understanding.

VR-based environments have been shown to reduce extraneous cognitive load by making spatial relationships directly perceivable, allowing learners to focus cognitive resources on germane processing. The primary motivation for designing an imaginary binary star system for this project, rather than replicating our known solar system, is to push learners out of familiar conceptual frameworks. By embedding the user inside a custom-designed environment featuring dual host stars (Solari and Lumina) and diverse, procedurally generated planets, the application requires them to actively construct new spatial intuitions. The VR application renders the complex three-dimensional geometry of a multi-star system immediately apparent, demonstrating varied atmospheric conditions and orbital mechanics without requiring effortful mental construction.

Several prior systems have explored VR in adjacent areas. Ynnerman et al. developed a public-science visualization of the solar system using large-format immersive displays. Yu and Sahami evaluated desktop-based virtual planetarium software for introductory astronomy courses. More recently, Parong and Mayer examined the conditions under which immersive VR benefits learning compared to slideshow instruction, finding that engagement and presence significantly modulated outcomes. However, a dedicated interactive VR system specifically targeting multi-star extrasolar comprehension—where users navigate hypothetical, diverse planetary biomes—remains underexplored in the educational design literature.

The present work addresses this gap by developing the VRSES as an exploratory design project. Rather than conducting a traditional comparative study, this paper focuses on the architectural and atmospheric challenges of simulating a binary system, demonstrating how procedural VR environments can serve as effective, scalable pedagogical tools for visualizing advanced astronomical concepts.

II. DEVELOPMENT METHODOLOGY

This section outlines the core objectives and the iterative development workflow used to build the VR-Based Space Exploration System (VRSES).

A. Project Objectives

The development of this project was guided by the following practical objectives:

- To design and construct an immersive VR environment that enables users to interactively explore a hypothetical binary star system.
- To implement procedural rendering techniques, custom shaders, and orbital mechanics that accurately reflect the spatial dynamics of a multi-body astronomical system.
- To apply architectural design principles that minimize extraneous cognitive load, keeping the user spatially oriented while navigating extreme environments like gas giants and storm worlds.

B. Development Workflow

The project adopts an iterative, Design-Based approach to build and refine the VRSES. Fig. ?? illustrates the development pipeline. The workflow follows a structured cycle: conceptual ideation of the binary system (Solari and Lumina) and its planets, 3D modeling and procedural scene construction, VR interaction scripting in C#, and visual optimization.

Rather than deploying a quantitative user study, the methodology focuses on overcoming specific technical and environmental design challenges. These included simulating thick volumetric fog for the gas giant Aelos, rendering highly reflective, HDRI-driven ocean surfaces for the storm world Nereus, and mathematically scripting synchronised 360-degree orbital motion that maintains a constant 180-degree separation between the binary stars. By prioritising these immersive design solutions, the final system serves as a robust proof-of-concept demonstrating how custom-built VR environments can facilitate embodied learning.

III. SYSTEM ARCHITECTURE AND IMPLEMENTATION

This section presents the development framework for VRSES, detailing the ideation phase, 3D modeling, procedural scene construction, VR interaction scripting, and the system's role as an educational tool.

A. Ideation

The VRSES is designed using a learner-centered approach, placing the student as an active explorer within a hypothetical extrasolar environment rather than a passive recipient of information. The learner-centered design philosophy prioritizes learner autonomy and self-directed inquiry, with the instructor acting as a facilitator. In the ideation phase, learning objectives were defined around the following competencies: understanding the relative scale of stars and planetary bodies; grasping the geometry of multi-body orbital paths; recognizing the diversity of extrasolar system architectures; and interpreting basic stellar and planetary properties.

To motivate engagement and exploration, the system is structured as a free-roam navigable environment rather than a linear tutorial. Rather than recreating a known solar system, the project conceptualized a custom binary system featuring dual host stars (Solari and Lumina) and a diverse array of

orbiting bodies based on existing Unity scene architecture. These include a scorched, iron-rich inner world (**Pyronis**), a dense, extreme-atmosphere terrestrial planet (**Sulvara**), an arid, rocky world (**Dravix**), a massive gas giant (**Zephyrus**), and a storm-driven ocean world (**Thalor**). The spatial fidelity of the representations—proportional scale, elliptical orbital inclinations, and the approximate color temperatures of the binary stars—was prioritized to ensure that users are exposed to extreme atmospheric conditions and complex orbital mechanics in a scientifically grounded, yet imaginary, manner.

B. 3D Modeling and Scene Construction

The celestial bodies in the simulation were developed using pre-existing 3D assets sourced from online repositories, including both standard model files and Blender-compatible (.blend) formats. These assets were further refined and adapted within Unity to ensure consistency in scale, appearance, and performance. In addition to manually sourced models, certain visual elements were generated or enhanced using AI-based content generation tools to improve realism and diversity in planetary features.

Surface and atmospheric textures applied to planetary bodies were collected from multiple external sources and integrated as layered materials to represent variations in terrain, cloud structures, and atmospheric effects. Distinctions between terrestrial planets and gas giants were achieved primarily through the selection and combination of these texture maps. Similarly, stellar objects were assigned emissive materials approximating different spectral classes, using visually representative color schemes rather than physically simulated radiative models.

Orbital paths and animations were implemented within Unity using its built-in components, with trajectories approximated using simplified orbital parameters. This approach emphasizes visual clarity and conceptual understanding over strict physical accuracy, while still maintaining a coherent representation of orbital motion.

The rendered view of an extrasolar system scene in the VRSES, showing a host star and multiple orbiting planetary bodies illustrated in Fig. 1.

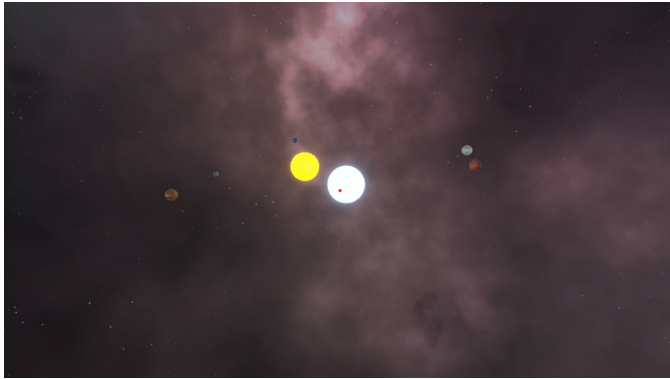


Fig. 1. Rendered view of an extrasolar system scene in the VRSES, showing a host star and multiple orbiting planetary bodies.

C. The Binary Extrasolar System and Planetary Environments

The core of the VRSES is built around a custom binary star system that drives the lighting and visual dynamics of the simulated universe. Orbiting these stars are five distinct planetary bodies, each engineered with specific environmental parameters, lighting profiles, and atmospheric rendering techniques to achieve targeted pedagogical and immersive goals.

I. The Host Stars: Solari and Lumina: The gravitational and illuminating anchors of the system are two host stars: **Solari** (The Yellow Star) and **Lumina** (The White Star). Solari functions as a main-sequence star, casting a warm, familiar golden light across the planetary bodies. In contrast, its companion, Lumina, is a hotter star that emits intense, cool-white radiation. The interplay of these two distinct light sources creates highly dynamic, double-cast shadows across the planetary surfaces. From the perspective of the VR user, looking up from any planet reveals the visually striking phenomenon of two suns traversing the sky, significantly enhancing the otherworldly immersion.

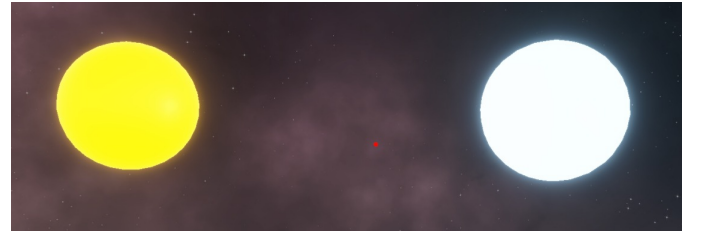


Fig. 2. View of the binary host stars, Solari and Lumina, illustrating the distinct color temperatures and illumination profiles of the dual-star system.

II. Pyronis: Pyronis is the closest planet to the binary star system. Lacking an atmosphere to distribute heat, the planet experiences extreme temperature variations. The terrain is entirely barren, rocky, and heavily cratered. Because there is no atmospheric scattering, the VR environment features extremely high-contrast lighting with incredibly sharp shadows. The complete absence of fog ensures clear, unobstructed visibility across the desolate landscape, emphasizing the harsh, unshielded exposure to stellar radiation.

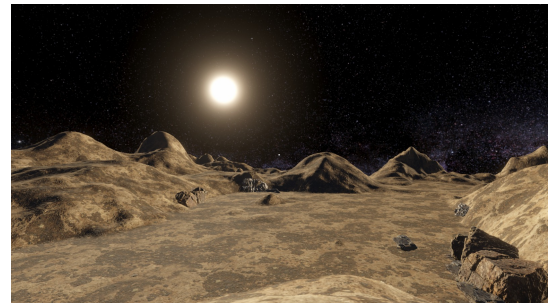


Fig. 3. Rendered VR view of the surface of Pyronis, illustrating high-contrast lighting, sharp shadows, and heavily cratered, barren terrain.

III. Sulvara: Sulvara is characterized by a thick, toxic sulfur dioxide (SO_2) atmosphere, resulting in crushing surface pressure and immense heat. The environment was constructed using Unity terrain tools to sculpt smooth, volcanic basalt rock and dusty soil, accented with emission materials to simulate glowing lava without the performance cost of real lights. A core feature of Sulvara is its dense exponential fog, which limits visibility to approximately 20–40 meters and creates a suffocating yellow-orange sky. Lighting is highly diffused, employing a warm directional light with disabled shadows to mimic atmospheric scattering. Post-processing effects such as bloom, lens distortion for heat waves, and point-light lightning flashes contribute to a hostile, dynamic VR experience.

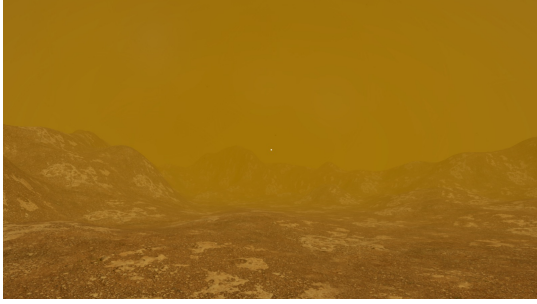


Fig. 4. First-person perspective on Sulvara, showing dense yellow-orange fog, volcanic terrain with emission lava, and diffused lighting effects.

IV. Dravix: Dravix represents a dry, iron-rich terrestrial world. The surface features sprawling red sand textures, deep craters, and rolling hills. To simulate its atmospheric conditions, a procedural skybox with an orange-red tone was combined with low-density exponential fog and a horizon glow to represent thin, dusty air. The environment includes particle systems simulating floating dust and wind movement. In VR, Dravix utilizes a custom physics simulation applying a reduced gravity of 3.7 m/s^2 , significantly altering player locomotion and falling mechanics. Invisible boundary colliders maintain immersion, while low-angle directional lighting and global volume bloom enhance the cinematic, desolate feel of the terrain.

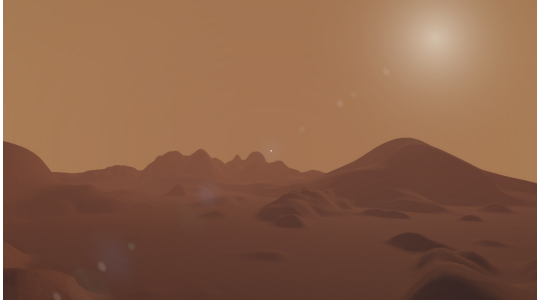


Fig. 5. Exploration of Dravix, highlighting the red dust terrain, low-angle lighting, and atmospheric dust particles.

V. Zephyrus: Unlike the terrestrial worlds, Zephyrus is a massive gas giant with no solid surface. To allow user

exploration, the VR environment places the player on a floating observation platform suspended deep within the thick atmosphere. The visual depth of the planet is achieved via a highly optimized multi-layer cloud system rather than computationally expensive true volumetrics. Layered transparent planes with specific sorting priorities, blending modes, and flat colors create the illusion of dense gas. Script-based UV offset animations apply continuous, varying wind speeds to these layers, bringing the atmosphere to life. A warm yellow fog blends the geometry, while massive, transparent ring structures loom in the sky above to provide planetary scale.

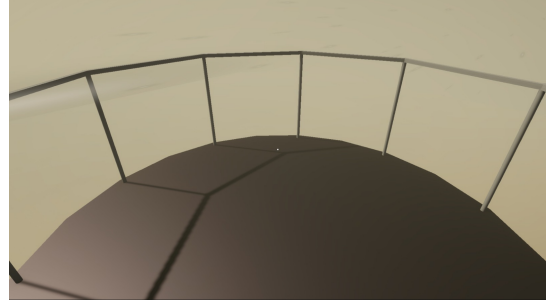


Fig. 6. View from the floating observation platform within the atmosphere of Zephyrus, showcasing multi-layered cloud rendering and planetary rings.

VI. Thalor: Thalor is a deep blue, extremely cold water giant dominated by strong rains and violent atmospheric storms. The visual profile relies heavily on a blue-toned environment integrated with dynamic atmospheric particle systems. The VR experience utilizes heavy rain visual and sound effects to simulate the torrential conditions of an active, storm-driven climate. Expansive bodies of water and a turbulent sky structure complete the immersive representation of this extreme aquatic planet.



Fig. 7. The storm-driven environment of Thalor, depicting deep blue atmospheric tones, heavy rain effects, and turbulent water bodies.

D. VR Interaction Development

The interaction layer of VRSES was implemented in the Unity 3D game engine using C# scripting. The system supports VR headset input via the Unity XR Interaction Toolkit, enabling deployment across standard PC hardware and consumer VR platforms. The key interaction modalities implemented are as follows.

Free Navigation: The user can fly through the scene using a standard first-person locomotion controller. Velocity scaling is implemented so that users can transition smoothly from a planetary-scale perspective (where individual planet surfaces and moons are visible) to a system-wide perspective (where the full orbital architecture of the star system is comprehensible in a single view).

Object Selection and Information Overlay: When the user directs their gaze or cursor toward a celestial object, an interactive information panel is triggered displaying the object’s name, type, key physical parameters (mass, radius, orbital period, distance from host star), and a brief descriptive summary.

System Selection Menu: A persistent UI panel allows the user to browse and select from the available catalogue of extra-solar systems. Upon selection, the scene transitions smoothly to the chosen system, recentring the camera at an appropriate viewing distance.

The control scripts are written in C# using the Unity editor. Table I summarizes the key technical specifications of the VRSES.

TABLE I
TECHNICAL SPECIFICATIONS OF THE VR-BASED SPACE EXPLORATION SYSTEM.

Specification	Details
Game Engine	Unity 3D (Unity 6000.4.0f1 version)
Programming Language	C#
Supported OS	Windows 10 / 11
VR SDK	Unity XR Interaction Toolkit
Exoplanet Data Source	NASA Exoplanet Archive
Modeling Reference	Kepler & TESS Mission Catalogues

E. VRSES as a Learning Tool

The VRSES is designed as an active, inquiry-driven learning environment for undergraduate students in physics, astronomy, and engineering. Unlike passive visualization tools, VRSES places the learner in direct control of their perspective and pace of exploration. Students can investigate the spatial layout of multiple extrasolar systems, compare the architectural diversity of different planetary systems, and develop an intuitive sense of astronomical scale that is difficult to acquire from textbook study alone.

IV. DISCUSSION AND CONCLUSION

The development of immersive planetary environments across multiple scenarios—namely *Pyronis*, *Sulvara*, *Dravix*, *Zephyrus*, and *Thalor*—demonstrates the effectiveness of virtual reality as a medium for simulating extreme and otherwise inaccessible extraterrestrial conditions. A central challenge addressed in this work was achieving perceptual realism under strict performance constraints inherent to VR systems. Rather than relying on computationally expensive volumetric simulations, the project adopted optimized techniques such

as exponential fog, layered transparent geometry, emission-based materials, and selective lighting models within the Unity engine using the Universal Render Pipeline (URP).

Across all environments, atmosphere simulation emerged as the most critical component in shaping user perception. For instance, the dense fog and diffused lighting in *Sulvara*, the multi-layered cloud structures in *Zephyrus*, and the thin, dust-based atmospheric effects in *Dravix* collectively illustrate how layered visual strategies can effectively replicate diverse planetary conditions. Similarly, the high-temperature, iron-rich surface characteristics of *Pyronis* and the deep, storm-driven atmospheric dynamics of *Thalor* further extend the range of environmental diversity achieved. These approaches align with established principles in perceptual rendering, where depth, scale, and realism are achieved through controlled visual cues rather than strict physical simulation.

Another significant contribution of this work lies in addressing rendering challenges associated with transparency and depth sorting—particularly in the multi-layered cloud system of *Zephyrus*. Through iterative debugging and techniques such as premultiplied blending, sorting priority adjustments, and spatial layer separation, stable and artifact-free rendering was achieved. This highlights the importance of low-level rendering considerations in building visually coherent VR systems.

From an interaction perspective, the integration of XR-based locomotion, gravity adaptation, spatial audio systems, and dynamic UI elements enhances user immersion and agency. These features transform the experience from passive visualization to active exploration, reinforcing the pedagogical and experiential potential of VR environments.

Overall, the project demonstrates that high levels of immersion can be achieved through the intelligent combination of simplified geometric representations, optimized rendering techniques, and user-centered interaction design, without the need for computationally intensive simulations.

This work presents the design and implementation of a suite of immersive VR-based planetary environments that simulate a diverse range of atmospheric and surface conditions. By leveraging optimized rendering strategies within Unity’s URP framework, the system successfully balances visual realism with real-time performance requirements essential for VR applications.

The resulting environments effectively capture key planetary characteristics, including dense atmospheric scattering, diffused lighting, layered cloud depth, and terrain-specific features, while maintaining smooth and responsive user interaction. The project further demonstrates that complex environmental phenomena can be convincingly approximated using lightweight, scalable techniques such as fog layering, emission-based materials, and procedural visual effects.

In conclusion, this research highlights the potential of virtual reality as a powerful tool for planetary visualization, immersive education, and experiential learning. The environments represented by *Pyronis*, *Sulvara*, *Dravix*, *Zephyrus*, and *Thalor* collectively showcase how diverse planetary conditions can be simulated efficiently while maintaining high levels of immer-

sion. Future work may extend this framework by incorporating real scientific datasets, enabling multi-user collaborative exploration, and integrating more advanced physically-based simulations to further enhance realism and educational value.

SOURCE CODE

The VRSES source code is available at: https://github.com/Kartiks5161/DES643_Project

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